

We propose a simple framework to think about **how long** it takes for **Transformers** to **learn** certain abilities: **sparse attention**.

We explain why:
1. Sparse attention **emerges**

2. Learning time increases as **sequence length increases** and as data gets more **diverse**.

3. **Repetition** can speed up learning.

Paper



ETH zürich

EPFL

Google DeepMind

The emergence of sparse attention: impact of data distribution and benefits of repetition

Nicolas Zucchet, Francesco D’Angelo, Andrew Lampinen, Stephanie Chan

Goals

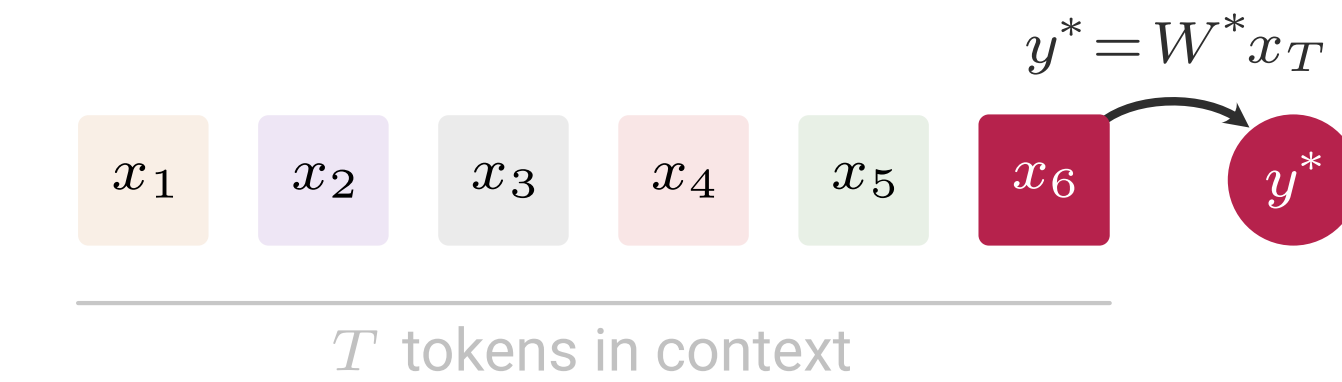
- Provide a **simple and general theory** to understand how long it takes to a Transformer to learn certain abilities, with a focus on the ones that emerge during learning (e.g., in-context learning)
- Understand the **role of data**, in particular why **repetition** has been found to speed up emergence empirically

Looking ahead

- Can we extend the theory to precisely understand the learning of **compositions of circuits** (induction heads + more)?
- Sparse attention is ubiquitous in LLMs, **more emergent abilities than what we currently know?**
- How far can we push **repetition** in practice?

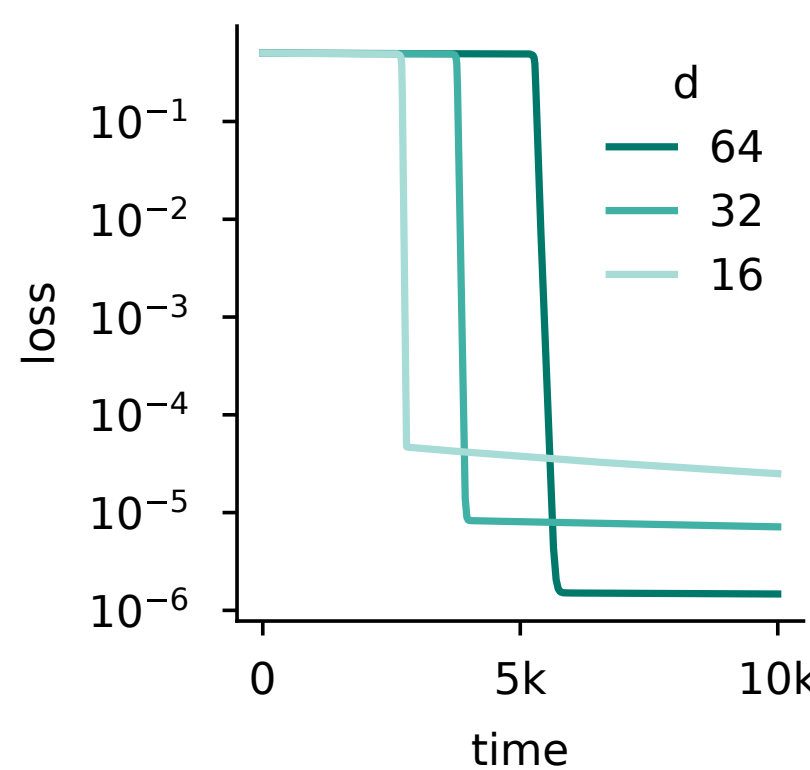
Attention emerges, in a single-location regression task

Task. Single-location linear regression

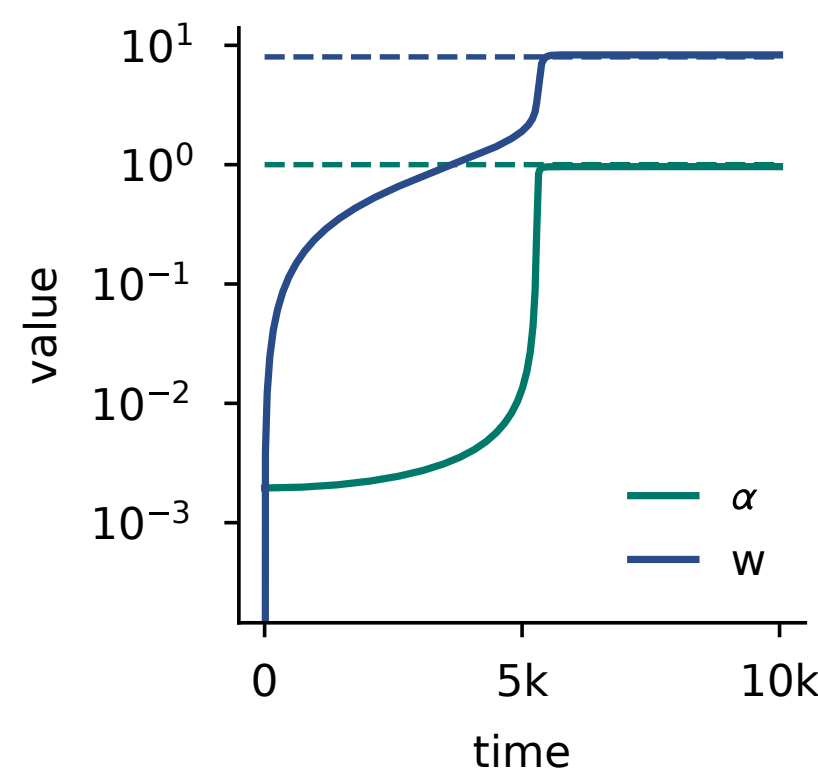


Model. Simplified Transformer

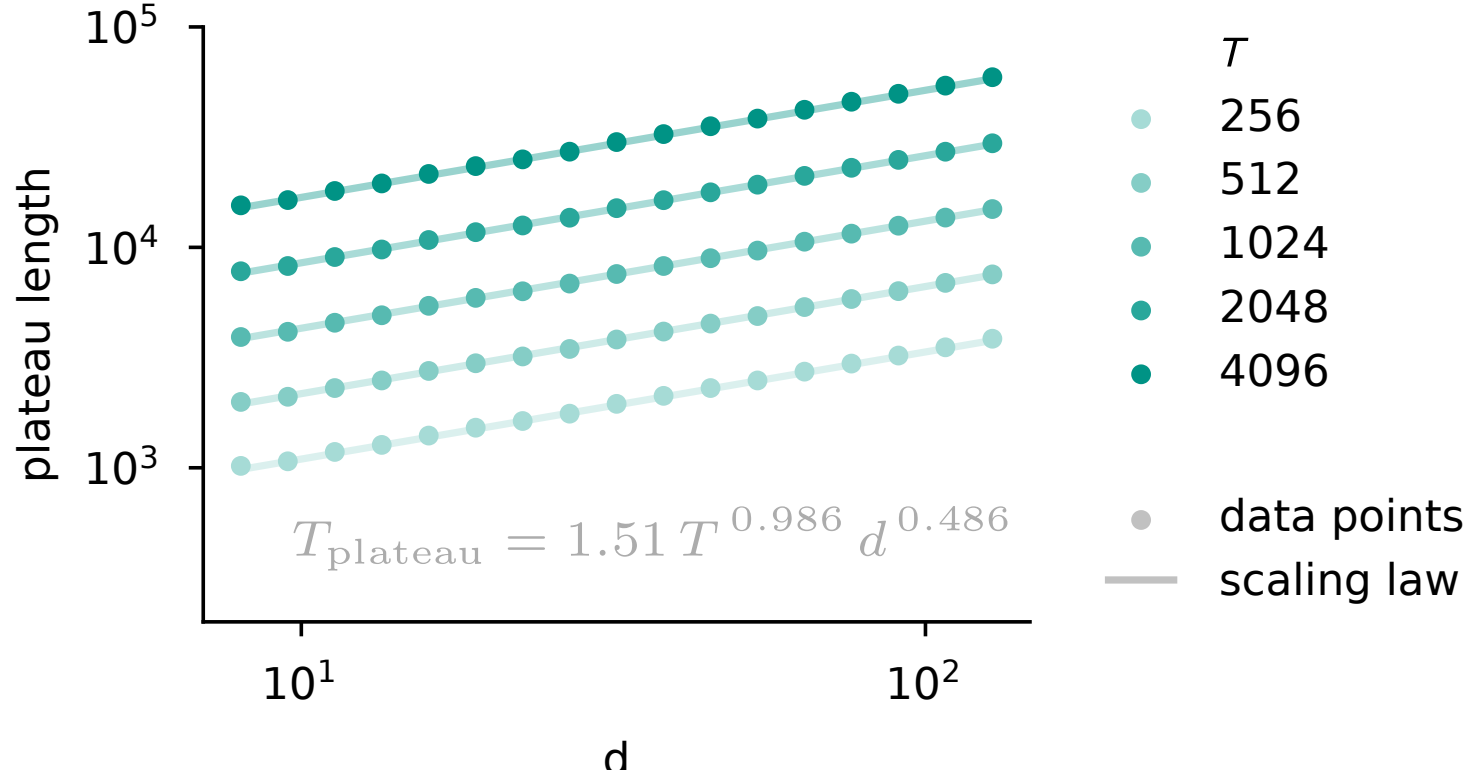
$$y = W \sum_{t=1}^T \text{softmax}(a)_t x_t \quad x, y \in \mathbb{R}^d$$



Exhibits emergent dynamics



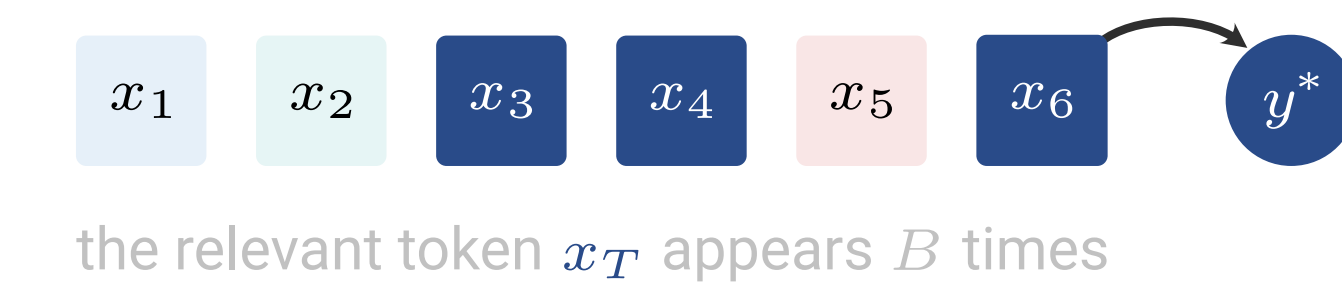
Feedforward weights W learn before attention formation



In theory: $T_{\text{plateau}} \sim \sqrt{dT}$

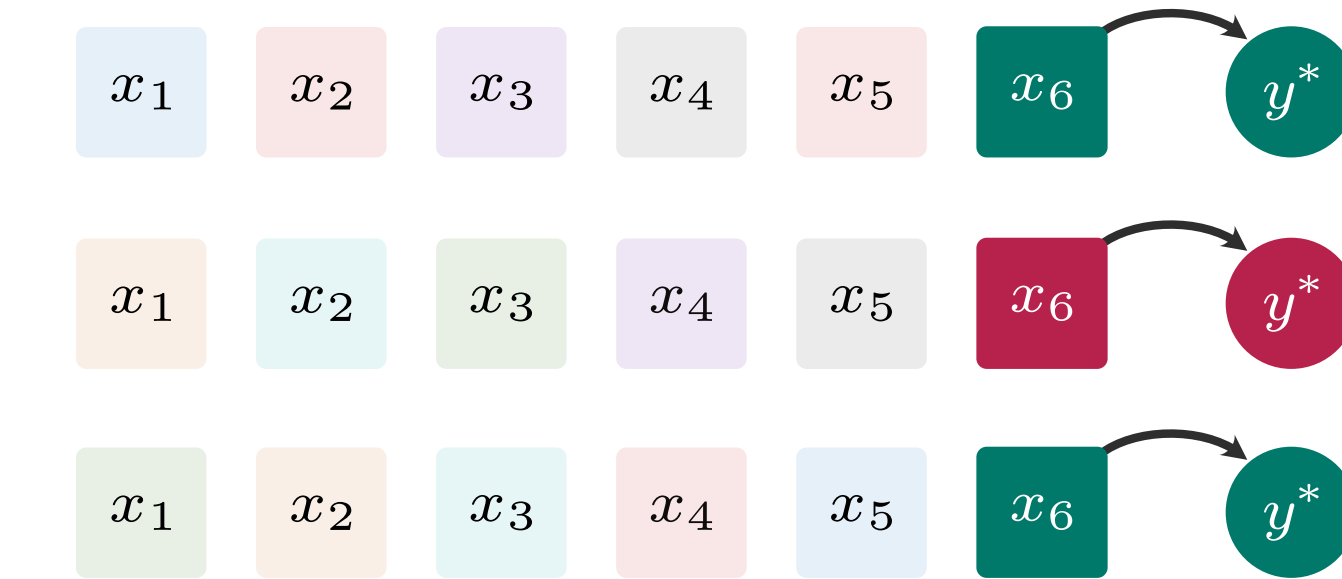
Repetition speeds up learning

In-context repetition



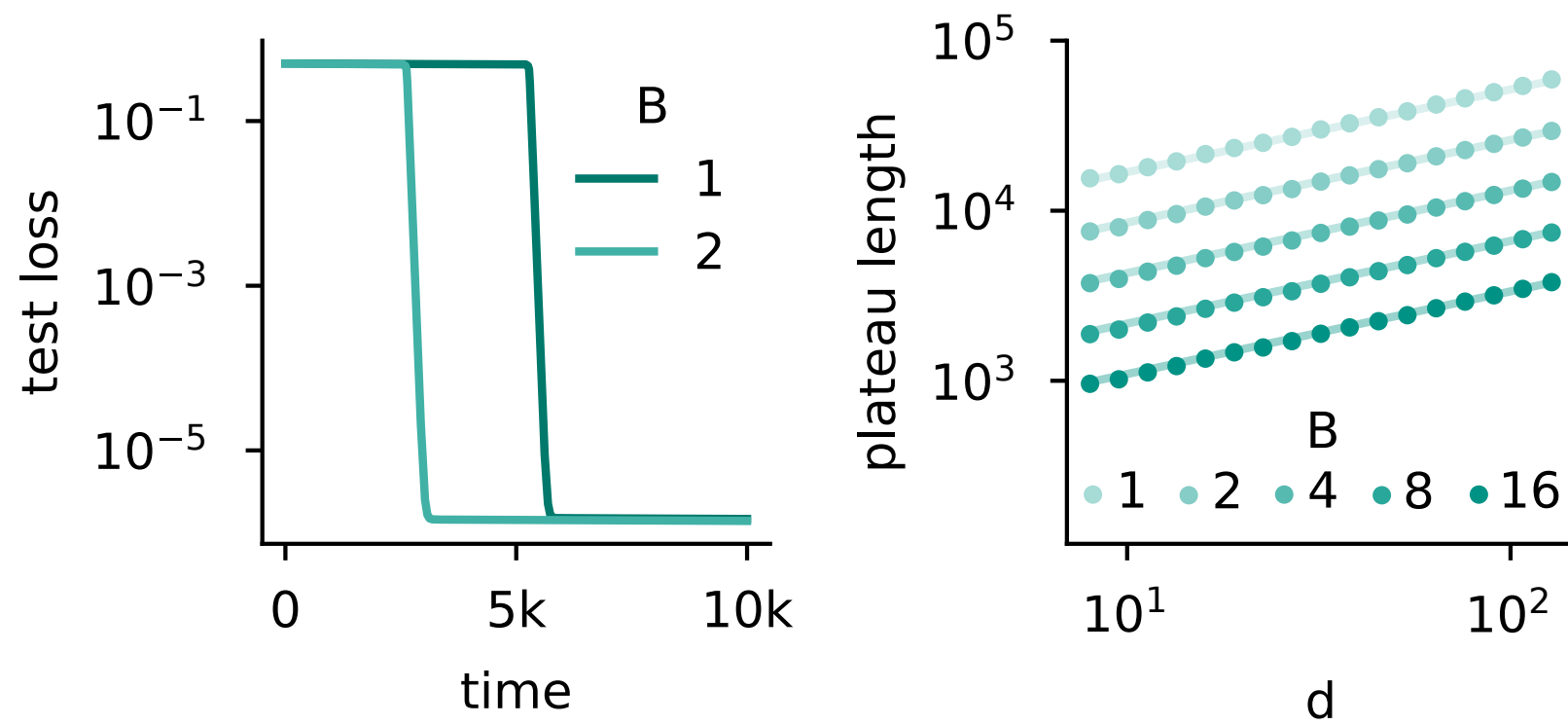
In a Harry Potter chapter, [Harry Potter] appears multiple times within the context

Cross-sample repetition



In Harry Potter books, [Harry Potter] appears in more chapters than [Sirius Black]

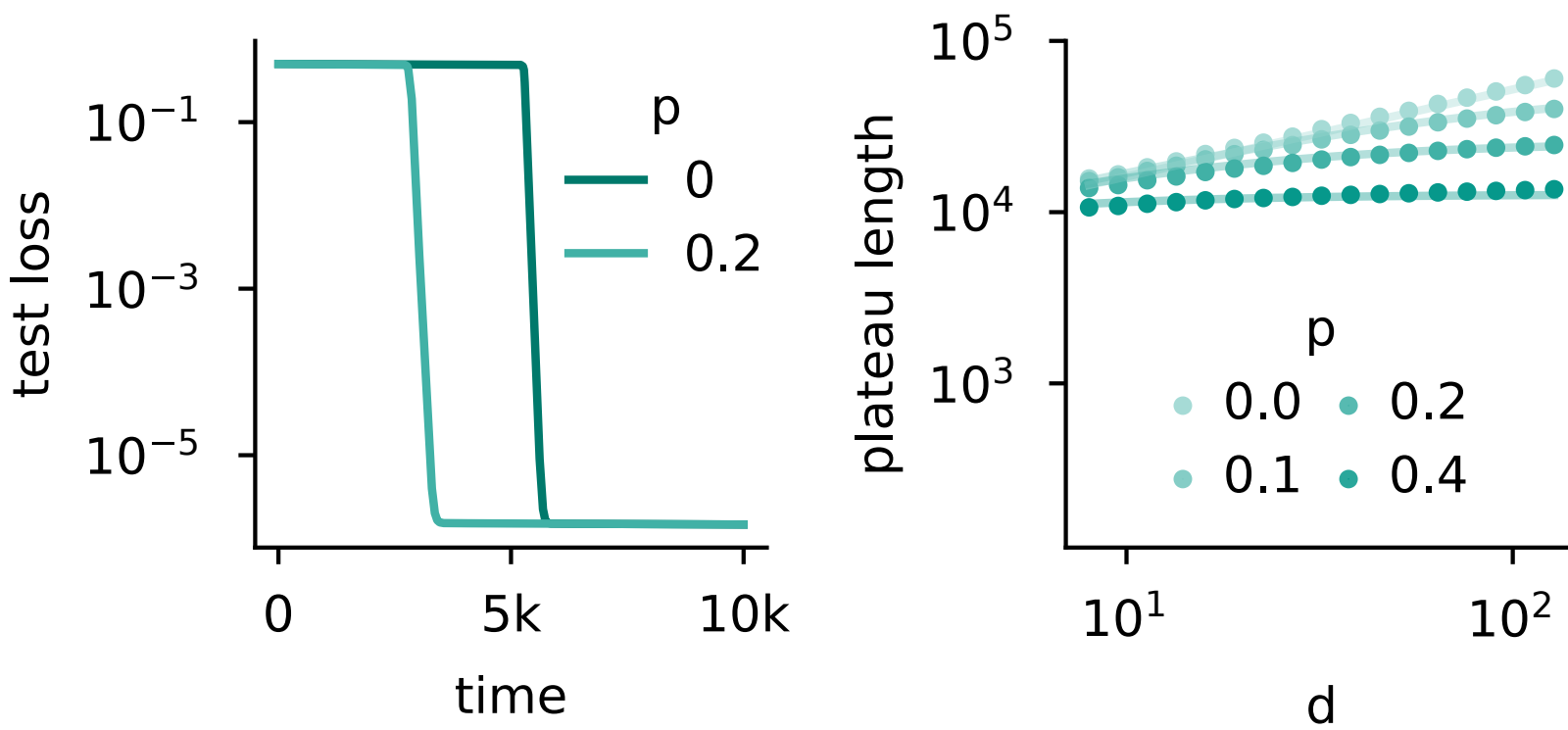
In-context repetition



In theory: $T_{\text{plateau}} \sim \sqrt{dT}/B$

Repetition **reduces** the effective **sequence length**

Cross-sample repetition

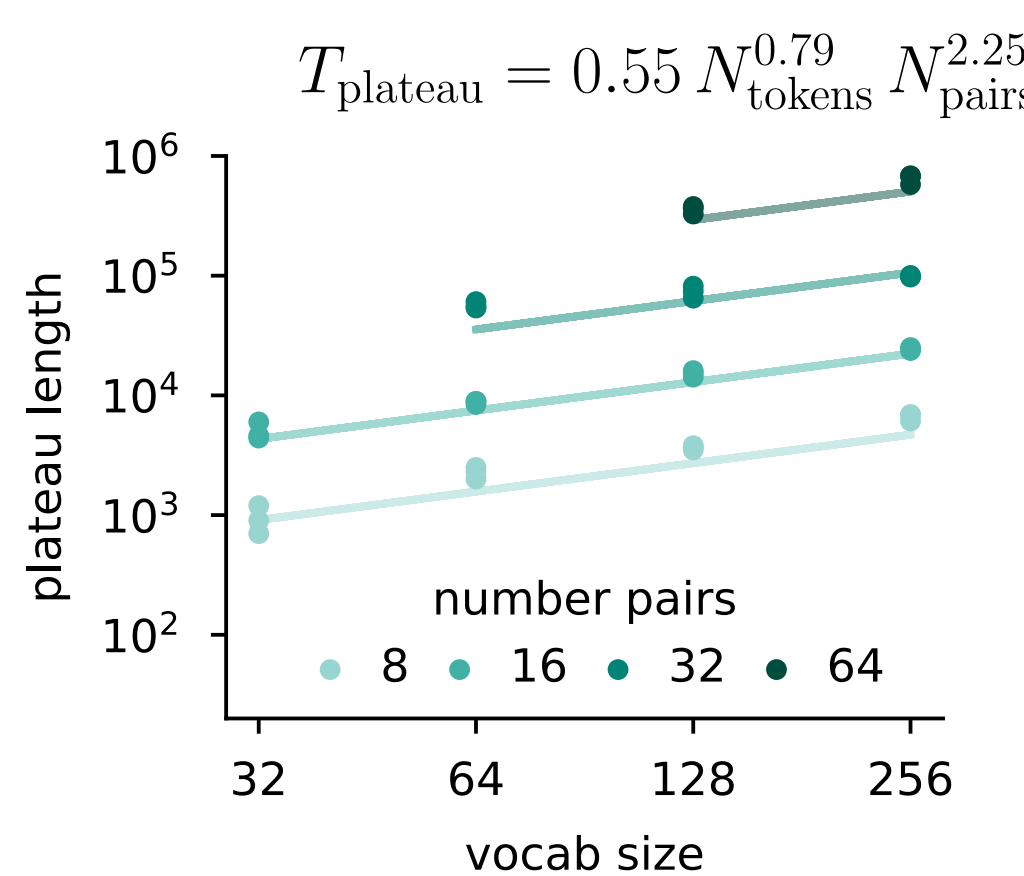
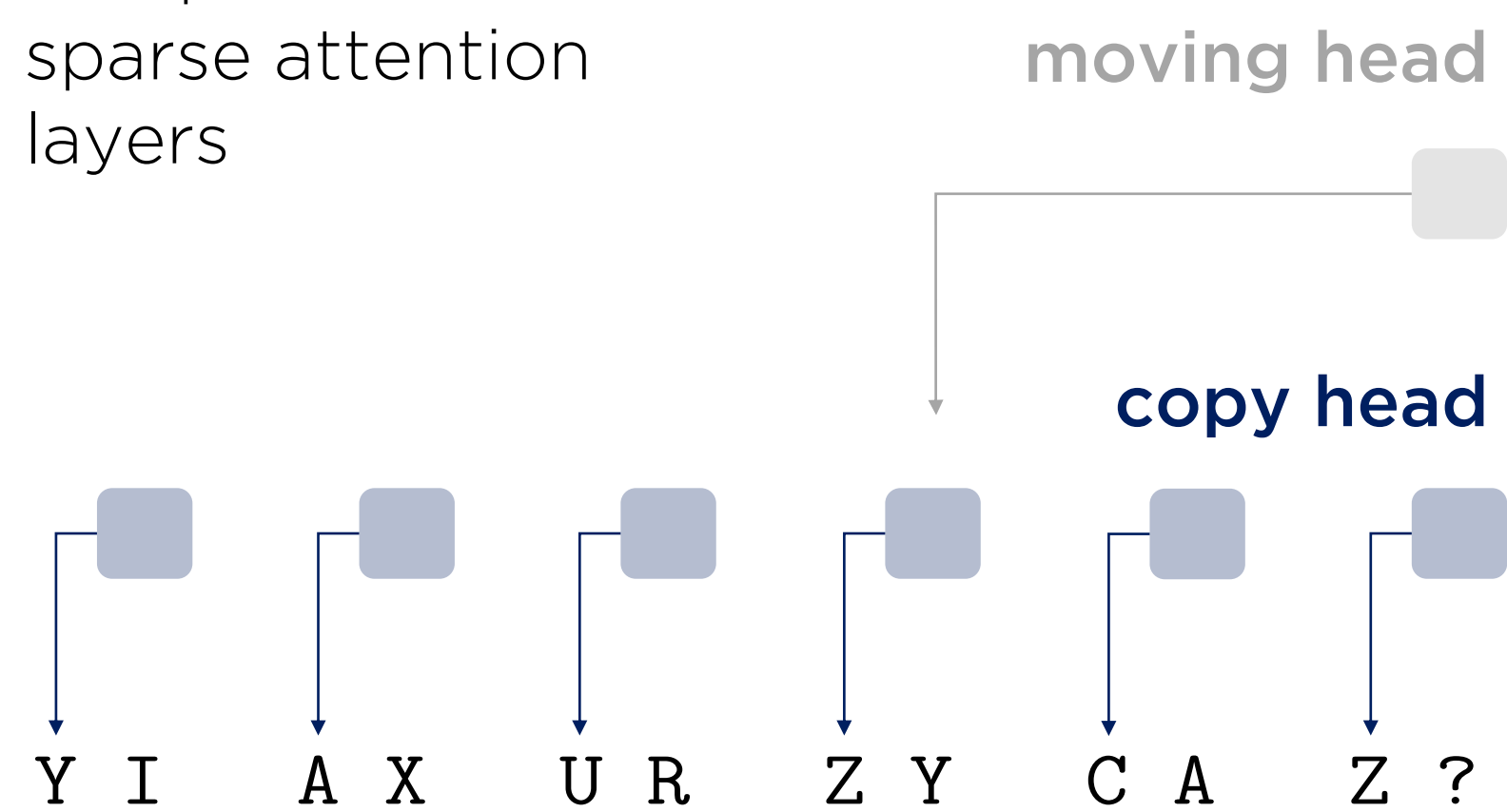


In theory: $T_{\text{plateau}} \sim \sqrt{dT}/\sqrt{p^2d + (1-p)^2}$

W learns faster on **repeated data**, which speeds up attention learning + **learning overall**

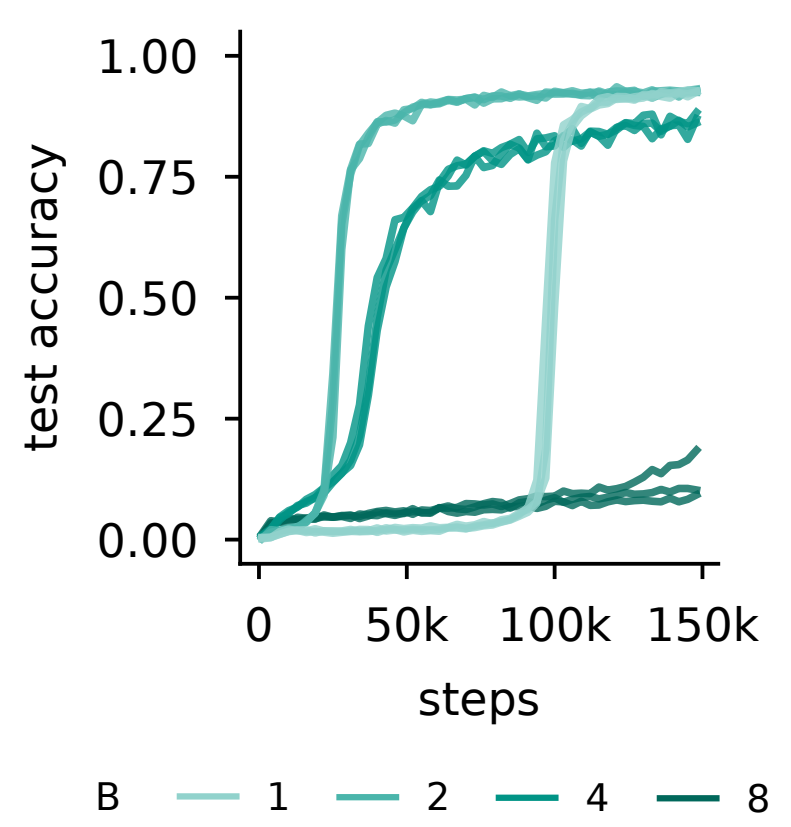
The sparse attention lens qualitatively explains the emergence of in-context learning

Induction head: composition of 2 sparse attention layers

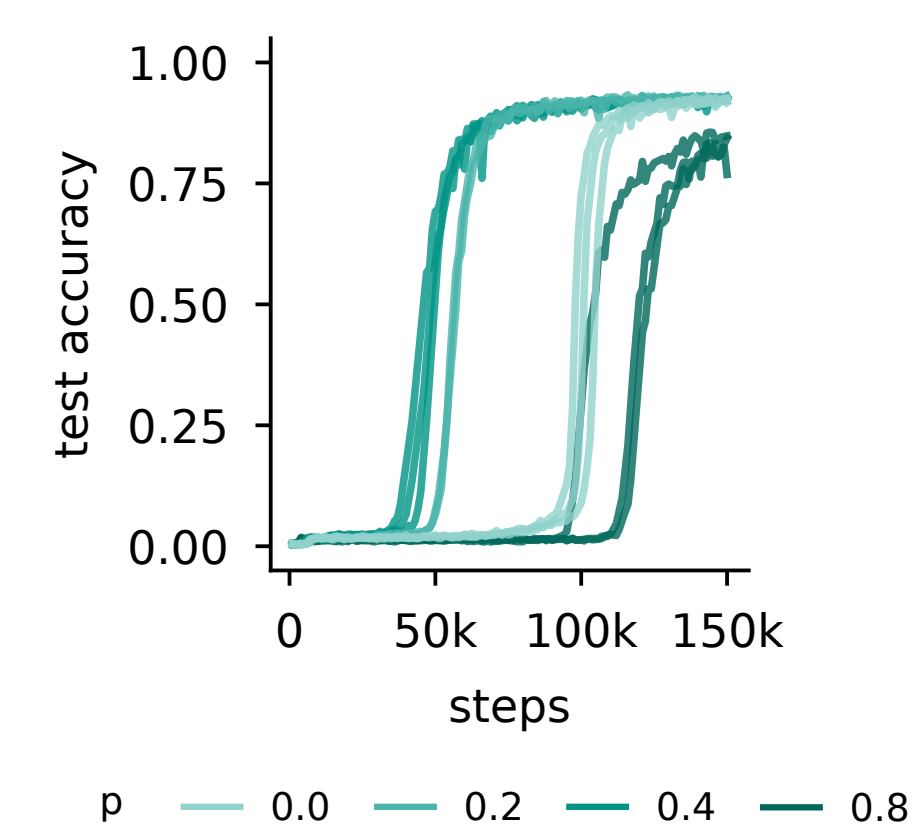


Same (qualitative) **behavior** as in the regression task

In-context repetition



Cross-sample repetition



Repetition **speeds up** training, but leads to **overfitting**